VECTORS · THE UNIFYING CONSTRUCTION

One construction, chosen coefficients

Every vector operation is $c_1\mathbf{v}_1 + \cdots + c_k\mathbf{v}_k$ with the coefficients fixed in advance. What separates the entries below is only which coefficients are allowed — pinned to specific values, or left free.

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FORMS

COEFFICIENTS PINNED — THE BASIC OPERATIONS 4

1 **Scalar multiplication**

one vector, one coefficient

$$c\mathbf{v}$$

The shortest combination there is. Its span is the line through $\mathbf{v}$, which is the $k = 1$ case of everything below — see basic operations.

2 **Addition**

both coefficients 1

$$1 \cdot \mathbf{u} + 1 \cdot \mathbf{v}$$

Addition is not a separate operation that combinations generalise — it is the combination with both coefficients set to one. The tip-to-tail picture is the geometric reading of exactly this.

3 **Subtraction**

coefficients 1 and −1

$$1 \cdot \mathbf{u} + (-1) \cdot \mathbf{v}$$

Likewise, with one coefficient negated. This is why subtraction needs no rules of its own: everything true of combinations applies to it directly.

4 **Trivial combination**

always available, always gives 0

$$0\mathbf{v}_1 + \cdots + 0\mathbf{v}_k = \mathbf{0}$$

Available for any set whatsoever, which is why finding **a** combination equal to $\mathbf{0}$ proves nothing. Independence asks whether it is the **only** one — the whole subject turns on that word.

COEFFICIENTS FREE — WHAT THE CONSTRUCTION GENERATES 3

5 **General combination**

arbitrary scalars

$$c_1\mathbf{v}_1 + \cdots + c_k\mathbf{v}_k$$

The definition the rest are instances of. Note it is a finite sum with arbitrary real coefficients — no convergence question arises, which is what separates this from an infinite series.

6 **Span**

the set of every reachable vector

$$\mathrm{Span}\{\mathbf{v}_i\} = \{\ \text{all combinations}\ \}$$

Let the coefficients range over everything and collect the results. The span is automatically a subspace — a combination of combinations is a combination — which is the cheapest subspace proof available.

7 **Membership as a system**

columns of A are the $\mathbf{v}_i$

$$\mathbf{b} \in \mathrm{Span} \iff A\mathbf{c} = \mathbf{b} \text{ solvable}$$

The bridge to computation. Asking whether $\mathbf{b}$ is a combination is asking whether a linear system is consistent, and the coefficients are its solution — which is why row reduction answers questions about spans.

The progression is one of freedom. Fix every coefficient and you get a single operation; free them and you get the span; ask whether a particular combination can reach $\mathbf{0}$ and you get independence. Nothing new is introduced at any stage — only the question asked about the same expression.